

Domain-Generalizable Phishing URL Detection under Realistic Evaluation Splits

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Abstract

Machine-learning phishing detectors often report near-perfect results on randomly divided URL collections, yet deployment requires recognizing attacks from domains absent during training. We define domain-generalizable phishing URL detection under leakage-resistant evaluation: classify an unseen URL when its registrable domain (effective top-level domain plus one, eTLD+1) is excluded from training. Comparing random, host-disjoint, eTLD+1-disjoint, and cross-source tests will quantify how much conventional evaluation overstates operational performance.

1 Introduction and Motivation

Phishing pages impersonate trusted services to steal credentials and financial information. The Anti-Phishing Working Group observed 3.8 million attacks in 2025 [2]. Because a URL can be inspected before navigation, reliable classification could protect browsers, email gateways, and messaging platforms. Prior work reports accurate machine-learning detection [3]; operational value, however, depends on recognizing new attacker domains without excessive false alarms.

1.1 Motivating Example

Consider a message asking the user to verify a PayPal account at <https://paypal.login-fraud.top>. Although the trusted brand appears on the left, the registrable domain is `login-fraud.top`. Flagging it as phishing before navigation could prevent credential disclosure, even when this domain never appeared in training.

2 Gap and Project Objective

Random splits can place related URLs in training and testing, making memorization resemble generalization. Cross-dataset transfer has reduced detector performance by 10–32% [4]. A 2026 protocol uses host-disjoint splits but reports 1,387 shared eTLD+1 domains between training and testing [1]; subdomains of one registrable domain can therefore cross the boundary.

We will treat eTLD+1 as the grouping unit and compare random-record, host-disjoint, eTLD+1-disjoint, and cross-source evaluation. Experiments will use the 235,795-site PhiUSIIL corpus [3] and an independent public URL corpus. We ask how estimated performance changes as leakage controls strengthen and whether conclusions persist across recall, precision, macro-F1, Matthews correlation, false-positive rate, and calibration. The contribution is a reproducible split protocol and an estimate of evaluation optimism.

3 Formal Problem Definition

Let $\mathcal{U}$ be the set of URL strings, $\mathcal{Y} = \{0, 1\}$, and $r : \mathcal{U} \rightarrow \mathcal{R}$ map a URL to its registrable domain (eTLD+1). Label $y = 1$ denotes phishing and $y = 0$ denotes legitimate. Given a labeled training corpus $D_{\text{tr}} = \{(u_i, y_i)\}_{i=1}^n \subset \mathcal{U} \times \mathcal{Y}$, the task is to learn a function

$$f : \mathcal{U} \rightarrow \mathcal{Y}$$

that predicts the label of each URL u drawn from a test distribution whose registered-domain support is disjoint from training:

$$r(u) \notin \{r(u_i) : (u_i, y_i) \in D_{\text{tr}}\}.$$

In cross-source evaluation, test URLs also come from a collection process absent from D_{tr} . The output is one Boolean label per URL; no model or feature representation is assumed.

3.1 Example Instance

For $u = \text{https://paypal.login-fraud.top}$, suppose `login-fraud.top` is absent from training. The expected output is $f(u) = 1$ (phishing).

4 Team and Collaboration

Prabhnoor leads provenance and split validation; Divraj leads benchmarking and reproducibility; Dilraj leads dataset integration, experiment tracking, repository maintenance, and error analysis. Their complementary data-engineering, ML, and cybersecurity experience motivates the team. All members contribute to formulation, review, implementation, experiments, analysis, and writing. Repository: <https://github.com/prabhnoob/domain-generalizable-phishing-url-detection>.

References

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